Part 2

Argument Essay

Topic: Is it necessary to control invasive species?

Directions:

Closely read and analyze each of the four sources provided on pages 11 through 18. Using evidence from at least ***three*** of these sources, plan and write a well-developed argument regarding whether it is necessary to control invasive species.

☑ **Be sure to:**

☐ Establish your claim regarding whether it is necessary to control invasive species.
☐ Distinguish your claim from alternate or opposing claims.
☐ Use specific, relevant, and sufficient evidence from at least ***three*** of the sources to support your argument.
☐ Cite your evidence by source number and line number(s) or graphic to avoid plagiarism. For example: (Source 1, lines 4–5), or (Source 2, graphic), or equivalent; do ***not*** simply summarize the sources.
☐ Organize your ideas in a cohesive and coherent essay.
☐ Maintain an appropriate style and tone, avoiding the use of casual or conversational language, symbols and/or abbreviations, unless quoting directly from a cited source.
☐ Use appropriate grammar and punctuation.

Note:

- Use scrap paper and the margins of this test booklet to take notes and plan your essay.
- Write your essay in ***black*** or ***blue ink*** beginning on page 1 of your essay booklet.

Source 1

"Bucking Convention to Track the Upside of Invasive Species"

A little over a decade ago, Jason Gleditsch was removing Asian honeysuckle when he noticed the birds. Robins and gray catbirds flocked around the thickets in autumn, attracted by the fat, ripe fruits. Originally introduced as ornamental plants in the early 20th century, Asian honeysuckle rapidly spread across the Eastern Seaboard of the United States, growing in dense stands that can shade out native plants. Conservationists often destroy these clumps. But Gleditsch, then an undergrad at Pennsylvania State University, noted something interesting: Native bird species seemed more drawn to the honeysuckle patches.

Intrigued, Gleditsch conducted a set of experiments — removing honeysuckle in some places, placing potted native plants in others, and counting birds. Not only did many native bird species adore honeysuckle, he wrote in a 2010 paper, they also consumed the fruits of native plants placed near the thickets at a noticeably higher rate, potentially helping increase the spread of their seeds.

The response was immediate. While some biologists were interested by the findings, Gleditsch said, others dismissed them as unimportant. And angry emails and handwritten letters arrived calling him a hack. "It was kind of difficult, to be honest," Gleditsch told *Undark* [magazine] in a recent interview. "And it showed how emotionally charged this topic can be."

Over the last century, biologists have warned of the dire consequences of introducing alien organisms [invasive species] into new ecosystems; a whole field of study, called invasion science, has extensively cataloged the resulting damage. When some researchers, like Gleditsch, have suggested that certain alien species may help ecosystems, they've been met with fierce debate from invasion scientists who worry that such studies are a distraction from real damage. Researchers on the other side of the debate, however, argue that the relentless focus on the damage risks missing equally important data, and could lead to management decisions that do more harm than good. …

The field of invasion science arose in the late 1950s, a time of growing awareness that ecosystems across the world were under serious threat. Introduced species were a kind of natural experiment, said Daniel Simberloff, a prominent invasion biologist now at the University of Tennessee, and the results didn't look good. Insects like the emerald ash borer and fungi like chestnut blight have devastated American forests. Mesquite took over South African rangeland. Rabbits and cats chewed up flora and fauna in the Australian bush, while feral hogs rooted through farmers' fields throughout the American South. …

But while the damage done by some invasives was clear, the notion that they were always an inherent threat to native ecosystems troubled some researchers, said Jens-Christian Svenning, a researcher with the Center for Biodiversity Dynamics in a Changing World at Denmark's Aarhus University. A few critics of the field saw a heavy, unthinking bias toward nativism[1] and, perhaps, xenophobia.[2] Others pointed out that ecosystems might be more receptive to some alien species than the field assumed, Svenning said, noting potential biases in how scientists measured damage, and asked whether the benefits alien species might offer were going unstudied. …

Minor changes to a new ecosystem — an alien plant attracting pollinators to nearby native plants, for example — might help individuals from a native species without boosting their population numbers. A moderate change, such as Pennsylvanian birds thriving on Asian

[1]nativism — favoring the interests of native-born or established inhabitants

[2]xenophobia — fear or prejudice of anything that is strange or foreign

honeysuckle fruit, helps grow a native species population. The much rarer major and massive changes occurred when aliens led to native species expanding their territory or being saved from extinction — for example, when introduced ladybugs helped to devour pests afflicting native trees on the island of St. Helena.

One issue the study[3] has run into, [University of Fribourg Invasion Biologist, Giovanni] Vimercati acknowledged, is a severe lack of data. In general, most introduced species haven't been subjected to close scrutiny. (There are only so many grad students willing to devote a decade to studying an animal that may or may not be a problem, and small things — from insects to microbes — often get short shrift.) And of the papers that do exist, some suggest benefits of invasive species without providing much supporting evidence.

But there's also a bit of unconscious bias at play, Vimercati said. For decades, invasion biologists have focused their attention on negative impacts, and as a result, accumulated a lot of data pointing in that direction. Now, he added, "we're reaching a moment where we're considering positive impacts that in the past were simply overlooked." …

—Asher Elbein
excerpted and adapted from "Bucking Convention to Track the Upside of Invasive Species"
https://undark.org, September 28, 2022

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[3]the study — Giovanni Vimercati and a team of invasion biologists gathered papers on over 100 alien organisms with reported positive impacts to their surroundings

Source 2

"Invasive Species"

An invasive species is an organism that is not indigenous, or native, to a particular area. Invasive species can cause great economic and environmental harm to the new area.

Not all non-native species are invasive. For example, most of the food crops grown in the United States, including popular varieties of wheat, tomatoes, and rice, are not native to the region.

To be invasive, a species must adapt to the new area easily. It must reproduce quickly. It must harm property, the economy, or the native plants and animals of the region. …

Introduced Species…

Introduced species multiply too quickly and become invasive. For example, in 1949, five cats were brought to Marion Island, a part of South Africa in the southern Indian Ocean. The cats were introduced as pest control for mice. By 1977, about 3,400 cats were living on the island, endangering the local bird population.

Other invasive species descended from pets that escaped or were released into the wild. Many people have released pet Burmese pythons into the Everglades, a swampy area of south Florida. The huge snakes can grow to 6 meters (20 feet) long. Pythons, native to the jungles of southeast Asia, have few natural predators in the Everglades. They feast on many local species, including white ibis and limpkin, two types of wading birds.

Invasive Species and the Local Environment

Many invasive species thrive because they outcompete native species for food. Bighead and silver carp are two large species of fish that escaped from fish farms in the 1990s and are now common in the Missouri River of North America. These fish feed on plankton, tiny organisms floating in the water. Many native fish species, such as paddlefish, also feed on plankton. The feeding cycle of the paddlefish is slower than that of the carp. There are now so many carp in the lower Missouri River that paddlefish do not have enough food. …

Many invasive species destroy habitat, the places where other plants and animals naturally live. Nutria are large rodents native to South America. Ranchers brought them to North America in the 1900s, hoping to raise them for their fur. Some nutria were released into the wild when the ranchers failed. Today, they are a major pest in the Gulf Coast and Chesapeake Bay regions of the United States. Nutria eat tall grasses and rushes.[1] These plants are vital to the regions' marshy wetlands. They provide food, nesting sites, and shelter for many organisms. They also help secure sediment and soil, preventing the erosion of land. Nutria destroy the area's food web and habitat by consuming the wetland grasses. …

Eradicating Invasive Species

Officials have used a variety of methods to try to eradicate, or get rid of, invasive species. The cats on Marion Island were infected with a virus, for instance.

Sometimes other species are introduced to help control an invasive species. In Australia, prickly pear cactus, which is native to the Americas, was growing out of control. The cactus was destroying rangeland, where ranchers raised livestock. The government brought in cactus moth caterpillars to eat the cactuses. The caterpillars are natural predators of the cactus.

Introducing insects can be dangerous, however. Sometimes, the insects also damage other plant species — they can become invasive species themselves. Chemicals have also

[1]rushes — several different species of flowering plants that grow cylindrical, stem-like leaves

been used to control invasive species, but they can sometimes harm noninvasive plants and animals.

45 Governments are working to educate the public about invasive species. For example, in the United States, international fishing vessels are warned to wash their boats before returning home. This prevents them from accidentally transporting zebra mussels or other species from one body of water to another. …

—Kim Rutledge, et al.
excerpted from "Invasive Species"
https://education.nationalgeographic.org
October 30, 2024

Source 3

Encyclopedia of Biological Invasions

As you read this, thousands of species of plants, animals, fungi, and microbes have been or are being transported by humans to new locations, whether deliberately or inadvertently. This geographic rearrangement of the Earth's biota[1] is one of the great global changes now underway. Of course, species have always managed to spread, even without human assistance, but much less often, much more slowly, and not nearly so far. Over the last 150 years, beginning with the advent of steamships and accelerating with air travel, the rate of movement of some organisms has increased manyfold. Any distance can be quickly spanned by a plane; a hitchhiking seed, spore, or insect can be transported from Asia to South America, or from Africa to Australia, in a day. Although many introduced species fail to establish populations or remain restricted to the immediate vicinity of the new sites they land in, others establish populations and invade new habitats, spreading widely and sometimes well beyond the initial point of introduction. Their interactions, both within native communities and with other introduced species, have been noticed by biologists and, increasingly, by governments and the lay[2] public.

Ecologists, evolutionists, economists, geneticists, agronomists,[3] fisheries and forestry scientists, and many others study biological invasions for a variety of reasons. Some invaders are enormously costly, damaging agriculture, forestry, fisheries, and other human endeavors as well as natural areas. Other invaders are pathogens that cause human, animal, or plant disease; yet others are vectors[4] that carry these pathogens. Many invaders depress populations of native species, and even threaten some with extinction, by preying on them, competing with them, hybridizing with them, infecting them with disease, or changing their habitat. The cost of biological invasions to the economy of the United States alone is estimated at over $100 billion annually. …

Biological invasions have attracted attention for other reasons. The variety of impacts of introduced species is astounding. Many invasions have such idiosyncratic[5] and bizarre effects that they cannot fail to arouse our curiosity simply as fascinating tales of natural history. For example, who would have predicted that introducing kokanee salmon to Flathead Lake, Montana, and, many years later, opossum shrimp to three nearby lakes would ultimately have led to population crashes of grizzly bears and bald eagles through a complicated chain reaction? Or that introducing myxoma virus to Great Britain to control introduced rabbit populations would have led to the extirpation[6] of the large blue butterfly there? Who would have suggested that introducing a particular grass species would lead to hybridization with a native congener,[7] subsequent polyploidization,[8] and origin of a new vigorous invasive species that would change entire intertidal ecosystems? Teasing apart such intriguing causal chains is a scientific accomplishment of the first order. But the variety and idiosyncrasy of invasion effects also challenges attempts of invasion biologists to produce general laws or rules and

[1]biota — biological makeup

[2]lay — untrained

[3]agronomists — agricultural scientists

[4]vectors — transmitters

[5]idiosyncratic — individualized

[6]extirpation — elimination

[7]congener — relative

[8]polyploidization — the multiplication of a species' chromosome set to create a new species

to be able to explain why some introductions have no major impacts, yet others lead to huge invasions. Being able to predict which species will fall in the latter category if introduced, and which in the former, is the elusive holy grail[9] of invasion biology. …

—Daniel Simberloff and Marcel Rejmánek
excerpted from *Encyclopedia of Biological Invasions*, 2011
University of California Press

[9]holy grail — something valuable that is searched for

Source 4

The New Wild: Why Invasive Species Will Be Nature's Salvation

…Excited language, replete[1] with military and xenophobic metaphors, has continued to feature in the everyday discourse of scientists investigating alien species, and it even appears in their research papers. It may not sound very dispassionate or scientific, but it reflects the assumption, widespread since [biologist, Charles] Elton, that foreign species are up to no good and that their being alien means their impacts can be assumed to be bad. [Elton established the theory of invasive species.] They are guilty until proved innocent. In scientific journals, where researchers normally strive to use neutral language, those who call themselves "invasion biologists" stand out. The subjects of their inquiries "explode" on arrival, killing, eradicating, assaulting, and decimating native species, while overrunning, flooding, and devastating their new habitats. They secure beachheads and fight battles. …

In the past quarter-century, "invasion biology" has become a distinct academic discipline, with its own journals, conferences, research centers, and gurus, such as Elton and [Daniel] Simberloff. But there is growing criticism of its narrow agenda and apparent myopic[2] focus on demonstrating the hypothesis that aliens are bad. The charge is that invasion biologists have shown systematic bias in their studies. They have started from the presumption that alien species are bad and sought out research topics that confirm their view. Like tabloid editors, they concentrate their studies on the nastiest and most sensational invaders. They have rarely been open to other interpretations and rarely investigate aliens with beneficial or more nuanced impacts on their surroundings. …

We need some new icons to represent benign aliens. One might be the dandelion (*Taraxacum officinale*), which seems always to have been an unremarked presence across Europe and Asia. It will grow just about anywhere there is sunlight. It traveled everywhere with European imperialists, who believed it had medicinal uses. The British took it to New England, the French to Canada, and the Spaniards to California and Mexico. It likes disturbed land of the kind humans specialize in creating. Its puffballs of seeds easily blow long distances. It provides nectar for bees. Not even the most determined bio-xenophobe has come up with any nasty habits that make the dandelion a bad neighbor. One website I stumbled upon (since deleted) warned that "you can probably find it on any block in America." Since it does little discernible harm, that sounds like good news. …

Behind the demonology, what is the truth? One of the most common charges against alien species is that they almost inevitably cause extinctions among their new hosts. They cause a decline in biodiversity. The theory behind this is rooted in ecological thinking about how natural ecosystems are well-oiled machines in which every native species inhabits a unique "niche" with a particular job to do, whether as prey or predator, pollinator, or processor of waste. The ecosystem is "saturated" with species, and there is usually little room for interlopers[3] to make much of a contribution. So if a new species takes root, then it will usually supplant[4] a native. Expelled from the system, that native may become extinct, at least locally. …

In the past decade or so, a new body of evidence has emerged suggesting that some of the most widely used statistics in the canon of invasion biology do not stand up. When Dov Sax of Brown University conducted a global study of species invasions on oceanic islands, he found

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[1]replete — complete

[2]myopic — shortsighted

[3]interlopers — intruders

[4]supplant — replace

that while the number of birds did sometimes go down after invasions, plant biodiversity usually rose. Typically, such islands now have twice as many plant species as before. That study almost instantly put Sax at the forefront of a quiet revolution in which a new generation of researchers questioned the demonizing of aliens by Simberloff and others. …

—Fred Pearce
excerpted and adapted from
The New Wild: Why Invasive Species Will Be Nature's Salvation, 2015
Beacon Press